Food Waste Monitoring (Case Study 53)

Problem Statement

Compare meals prepared with meals served and calculate the number and percentage of meals wasted.

Objectives

Use lists, loops, arithmetic, and percentage calculations.

Assumptions

Meals prepared is greater than zero, and meals served cannot exceed meals prepared.

Algorithm

1. Read prepared and served meals for three days. 2. Calculate leftover meals. 3. Calculate waste percentage. 4. Report
whether waste is above 20%.

Flowchart (Text Form)

START → Read prepared and served counts → Calculate leftovers and waste % → Check threshold → Print daily report
→ STOP.

Pseudocode

Python Source Code

```
prepared_list = [100, 120, 90]
served_list = [85, 100, 80]
for i in range(len(prepared_list)):
    prepared = prepared_list[i]
    served = served_list[i]
    waste = prepared - served
    percent = waste / prepared * 100
    print("Day", i + 1, "leftovers:", waste)
    print("Waste percentage:", round(percent, 2), "%")
    if percent &gt; 20:
        print("High waste")
```

Test Case and Expected Output

Day 1: prepared=100, served=85
Expected: leftovers 15; waste percentage 15.0%; no 'High waste' message

Conclusion

The program measures leftover food and highlights days when waste exceeds the chosen limit.